The Covariant Formalism of Electromagnetism

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1 Motivation

The covariant formalism of classical electromagnetism is often never covered in bachelors(undergraduate) physics, despite its tendency to compactify Maxwell's Laws. Students often encounter it in particle physics or general relativity, as a demonstration of the usefulness of casting equations in tensorial form. This pedantic work is an attempt to bridge the gap between undergraduate special relativity and electromagnetism, and the graduate level presentation of the same topic. Understanding this formalism is very crucial, since the concepts of gauge, tensor manipulation and deriving the field equations from a Lagrangian inevitably crop up in advanced physics.

2 Convention

We will work in the framework of special relativity. This means that the indices are raised and lowered using the Minkowski metric, given by $\eta = diag(-1, 1, 1, 1)$. The partial derivative vector is $\partial_\mu = (\partial_t, \nabla)$, and its contravariant form is $\partial^\mu = (-\partial_t, \nabla)$.
IMPORTANT: The convention of the Minkowski metric adopted is the usual for general relativity. Particle physicists usually use a reversed signature.

3 Field Theory Essentials

We often use the word 'field' to describe electricity and magnetism. Mathematically, a field is a function that returns a value at every point, a map from points in space and time to a number of physical significance. The common method to derive the equations that these fields follow is to use the principle of least action. Construct the action,

$$S = \int \mathcal{L}(\phi, \partial_\mu \phi) d^4x, \tag{1}$$

$$\delta S = \int \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right) d^4x = \int \left(\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu(\delta\phi) \right) d^4x, \tag{2}$$

and imposing vanishing change of the fields in at the endpoints of spacetime, plus the usual partial integration gives us for a fields,

$$\frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} = 0, \tag{3}$$

the Euler Lagrange Equations. Now seeking conditions on the invariance of the Lagrangian will give us the field theory version of the stress-energy tensor. Again writing out the change in the Lagrangian under a transformation as,

$$\delta L = \frac{\partial \mathcal{L}}{\partial \phi_a}\delta\phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\delta(\partial_\mu \phi_a), \tag{4}$$

and using (3) that was derived before, the first partial derivative can be rewritten to give,

$$\delta L = \partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\delta\phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\partial_\mu(\delta\phi_a) = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\delta\phi_a \right). \tag{5}$$

Now we will replace $\delta\mathcal{L}$ and $\delta\phi_a$ as follows,

$$\delta\mathcal{L} = \epsilon^\nu \partial_\nu \mathcal{L}, \quad \delta\phi_a = \epsilon^\nu \partial_\nu \phi_a, \tag{6}$$

where ϵ^ν is a coordinate independent change in the spacetime coordinates. The two results give us,

$$\epsilon_\nu \partial^\nu \mathcal{L} = \epsilon_\nu \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\partial^\nu \phi_a \right), \tag{7}$$

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\partial^\nu \phi_a \right) - \partial^\nu \mathcal{L} = 0, \tag{8}$$

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\partial^\nu \phi_a \right) - \eta^{\nu\mu}\partial_\mu \mathcal{L} = 0, \tag{9}$$

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\partial^\nu \phi_a - \eta^{\nu\mu}\mathcal{L} \right) = 0 \implies \partial_\mu T^{\mu\nu} = 0. \tag{10}$$

Adding any divergence free term to the stress energy tensor $T^{\mu\nu}$ will keep the whole term divergence free. The work of this section will allow us to describe the whole covariant formalism of electromagnetism ahead.

4 Maxwell's Equations and the Continuity Equation

The inhomogeneous Maxwell's Equations are,

$$\nabla \cdot \vec{E} = \rho, \tag{11}$$

$$\nabla \times \vec{B} = \frac{\partial \vec{E}}{\partial t} + \vec{J}, \tag{12}$$

and the homogeneous equations are,

$$\nabla \cdot \vec{B} = 0, \tag{13}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}. \tag{14}$$

The continuity equation for the sources can be established by taking the divergence of the second inhomogeneous equation as follows,

$$\nabla \cdot (\nabla \times \vec{B}) = \nabla \cdot \frac{\partial \vec{E}}{\partial t} + \nabla \cdot \vec{J}, \tag{15}$$

$$0 = \partial_t \rho + \nabla \cdot \vec{J} = \partial_\mu j^\mu, \quad j^\mu = (\rho, \vec{J}). \tag{16}$$

5 Gauge Freedom

We now want to replace $\vec{E}$ and $\vec{B}$ by some potential field with four components, A^μ, where $\mu = 0, 1, 2, 3$ to express the same four laws. Consider the following,

$$A^\mu = (\phi, \vec{A}), \quad \vec{E} = -\nabla\phi - \frac{\partial \vec{A}}{\partial t}, \quad \vec{B} = \nabla \times \vec{A}. \tag{17}$$

First we will note that the four vector A^μ can be shifted by some quantity $-\partial^\mu \chi$ and the electric and magnetic fields stay the same. That is,

$$A^\mu \to A^\mu - \partial^\mu \chi, \tag{18}$$

would produce no change in the physical fields $\vec{E}$ or $\vec{B}$. Explicitly, we can see this manifest first in the electric field as,

$$-\nabla\phi' - \partial_t \vec{A}' = -\nabla(\phi + \partial_t \chi) - \partial_t(\vec{A} - \nabla\chi) = \vec{E}, \tag{19}$$

and similarly for the magnetic field as,

$$\nabla \times \vec{A}' = \nabla \times (\vec{A} - \nabla\chi) = \vec{B}. \tag{20}$$

6 Deriving the Wave Equation in the Lorenz Gauge

Now by imposing that the electric and magnetic fields are derived from some potential four vector as given by (17), we can ask what conditions should A^μ obey. First we will see that the homogeneous Maxwell equations are solved automatically as follows,

$$\nabla \cdot B = \nabla \cdot (\nabla \times \vec{A}) = 0, \tag{21}$$

$$\nabla \times \vec{E} = (\nabla \times -\nabla\phi) + (\nabla \times -\frac{\partial \vec{A}}{\partial t}) = 0 - \frac{\partial \vec{B}}{\partial t} = -\frac{\partial \vec{B}}{\partial t}. \tag{22}$$

Now we will see how the inhomogeneous equations give us a wave equation for A^μ in a specific gauge, the Lorenz gauge. Start with the inhomogeneous equation for the electric field,

$$\nabla \cdot \vec{E} = \rho, \tag{23}$$

$$\nabla \cdot (-\nabla\phi - \frac{\partial \vec{A}}{\partial t}) = \rho, \tag{24}$$

$$-\Delta\phi - \partial_t \nabla \cdot \vec{A} = \rho. \tag{25}$$

Similarly, the inhomogeneous equation for the magnetic field gives,

$$\nabla \times \vec{B} = \partial_t \vec{E} + \vec{J}, \tag{26}$$

$$\nabla \times (\nabla \times \vec{A}) = \partial_t(-\nabla\phi - \partial_t \vec{A}) + \vec{J}, \tag{27}$$

$$-\Delta\vec{A} + \nabla(\nabla \cdot \vec{A}) = -\partial_t \nabla\phi - \partial_t^2 \vec{A} + \vec{J}, \tag{28}$$

$$-\Delta\vec{A} + \partial_t^2 \vec{A} = -\partial_t \nabla\phi - \nabla(\nabla \cdot \vec{A}) + \vec{J}, \tag{29}$$

$$-\Delta\vec{A} + \partial_t^2 \vec{A} = -\nabla\left(\partial_t\phi + \nabla \cdot \vec{A}\right) + \vec{J}. \tag{30}$$

Now we fix the gauge, by choosing to work with only that A^μ whose components satisfy the so-called Lorenz gauge condition,

$$\partial_\mu A^\mu = \partial_t\phi + \nabla \cdot \vec{A} = 0. \tag{31}$$

This reduces the two inhomogeneous equations to the following,

$$-\Delta\phi + \partial_t^2\phi = \rho, \quad -\Delta\vec{A} + \partial_t^2 \vec{A} = \vec{J}, \tag{32}$$

which we can collectively write in the four-vector notation as,

$$-\Box A^\mu = j^\mu, \quad \Box = -\partial_t^2 + \Delta. \tag{33}$$

In the absence of source charges, the equation reduces to the simple wave equation for the field components A^μ.

$$\Box A^\mu = 0. \tag{34}$$

7 Staying in the Lorenz Gauge

So we made the specification in (31), but what must happen to keep the field components A^μ within this gauge? The key lies in the fact that there is freedom encoded in the function χ, as given by (18). Suppose we have that $\partial_\mu A^\mu = 0$, and we want to maintain this in $\partial_\mu C^\mu$, where $C^\mu = A^\mu - \partial^\mu \chi$. We can extract the condition on χ as follows,

$$0 = \partial_\mu C^\mu = \partial_\mu (A^\mu - \partial^\mu \chi) = \partial_\mu A^\mu - \Box \chi \implies \Box \chi = 0. \tag{35}$$

So the gauge fixing function χ must be harmonic.

8 The Faraday Tensor and Special Relativity

We have thus far written the Maxwell's equations in terms of a potential four vector. We can alternatively write it in terms of a second order antisymmetric tensor. This is the Faraday tensor(also called field strength tensor), $F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha$, which is quite clearly antisymmetric. Let us work out its components.

$$F_{\alpha\alpha} = 0, \quad \alpha = 0, 1, 2, 3. \tag{36}$$

$$F_{10} = \partial_x A_t - \partial_t A_x = -\partial_x \phi - \partial_t A^x = E^x, \tag{37}$$

$$F_{20} = \partial_y A_t - \partial_t A_y = -\partial_y \phi - \partial_t A^y = E^y, \tag{38}$$

$$F_{30} = \partial_z A_t - \partial_t A_z = -\partial_z \phi - \partial_t A^z = E^z, \tag{39}$$

$$F_{21} = \partial_y A_x - \partial_x A_y = \partial_y A^x - \partial_x A^y = -B^z, \tag{40}$$

$$F_{31} = \partial_z A_x - \partial_x A_z = \partial_z A^x - \partial_x A^z = B^y, \tag{41}$$

$$F_{32} = \partial_z A_y - \partial_y A_z = \partial_z A^y - \partial_y A^z = -B^x. \tag{42}$$

The matrix representation of the same is (where the first index is row, second index is column),

$$F_{\alpha\beta} = \begin{pmatrix} 0 & -E^x & -E^y & E^z \\ E^x & 0 & B^z & -B^y \\ E^y & -B^z & 0 & B^x \\ E^z & B^y & -B^x & 0 \end{pmatrix}, \tag{43}$$

and since tensors must transform as tensors via multivariable chain rule, so must the components of $F_{\alpha\beta}$. This means that different observers will measure different electric and magnetic fields. Classical electromagnetic theory is compatible with special relativity.

9 Compact Version of EOM

We can now make use of the Faraday tensor to write the Maxwell equations in an even more compact form. The inhomogeneous equations can be written in the format,

$$\partial^\beta F_{\alpha\beta} = j_\alpha. \tag{44}$$

We could manually check that the above satisfies the inhomogeneous equations, but a more elegant method would be to proceed as follows,

$$\partial^\beta \left(\partial_\alpha A_\beta - \partial_\beta A_\alpha\right) = j_\alpha, \tag{45}$$

$$-\Box A_\alpha + \partial_\alpha(\partial_\beta A^\beta) = j_\alpha. \tag{46}$$

This is exactly the type of equation we obtained while deriving the wave form equation for the potential four vector A^μ. Fixing the gauge to be the Lorenz gauge and setting the source charges to vanish gives back (34). The homogeneous equations on the other hand are captured by,

$$\partial_\alpha F_{\beta\gamma} + \partial_\beta F_{\gamma\alpha} + \partial_\gamma F_{\alpha\beta} = 0. \tag{47}$$

Picking $\alpha = 0, \beta = 1, \gamma = 2$,

$$= \partial_t F_{12} + \partial_x F_{20} + \partial_y F_{01}, \tag{48}$$

$$= \partial_t B^z + \partial_x E^y - \partial_y E^x, \tag{49}$$

$$= \partial_t B^z + \left(\nabla \times \vec{E}\right)^z = 0. \tag{50}$$

Picking $\alpha = 1, \beta = 2, \gamma = 3$,

$$= \partial_x F_{23} + \partial_y F_{31} + \partial_z F_{12}, \tag{51}$$

$$= \partial_x B^x + \partial_y B^y - \partial_z B^z, \tag{52}$$

$$= \nabla \cdot \vec{B} = 0. \tag{53}$$

Picking $\alpha = 2, \beta = 3, \gamma = 0$,

$$= \partial_y F_{30} + \partial_z F_{02} + \partial_t F_{23}, \tag{54}$$

$$= \partial_y E^z - \partial_z E^y + \partial_t B^x, \tag{55}$$

$$= \partial_t B^x + \left(\nabla \times \vec{E}\right)^x = 0. \tag{56}$$

Picking $\alpha = 3, \beta = 0, \gamma = 1$,

$$= \partial_z F_{01} + \partial_t F_{13} + \partial_x F_{30}, \tag{57}$$

$$= -\partial_z E^x - \partial_t B^y + \partial_x E^z, \tag{58}$$

$$= -\partial_t B^y - \left(\nabla \times \vec{E}\right)^y = 0. \tag{59}$$

10 The EOM from the Action Principle

We saw that (44) seamlessly reproduced the inhomogeneous Maxwell equations. Often in physics when we have a field equation, we aim to work backwards and ask if it is the stationary point of some functional. This is a topic in the calculus of variations and in physics is the so-called action principle. Consider the action functional,

$$\mathcal{S} = \int \left(-\frac{1}{4} F^{\mu\nu} F_{\mu\nu} - j^\mu A_\mu\right) d^4x = \int \left(\int \mathcal{L} d^3x\right) dt, \tag{60}$$

where the Lagrangian density is given by,

$$\mathcal{L} = -\frac{1}{4} F^{\alpha\beta} F_{\alpha\beta} - j^\alpha A_\alpha. \tag{61}$$

We can insert the definition of the Faraday Tensor to obtain,

$$\mathcal{L} = -\frac{1}{4} \left(\partial^\alpha A^\beta - \partial^\beta A^\alpha\right) \left(\partial_\alpha A_\beta - \partial_\beta A_\alpha\right) - j^\alpha A_\alpha, \tag{62}$$

$$\mathcal{L} = -\frac{1}{4}\left(\partial^\alpha A^\beta \partial_\alpha A_\beta - \partial^\alpha A^\beta \partial_\beta A_\alpha - \partial^\beta A^\alpha \partial_\alpha A_\beta + \partial^\beta A^\alpha \partial_\beta A_\alpha\right) - j^\alpha A_\alpha. \quad (63)$$

In the parenthesis, the middle two terms and the outermost terms are the same, since all the indices are just dummy indices.

$$\mathcal{L} = -\frac{1}{2}\left(\partial^\alpha A^\beta \partial_\alpha A_\beta - \partial^\alpha A^\beta \partial_\beta A_\alpha\right) - j^\alpha A_\alpha. \quad (64)$$

The Euler-Lagrange equations for the fields A_μ are given readily as,

$$\frac{\partial \mathcal{L}}{\partial A_\nu} = j^\nu. \quad (65)$$

The first term requires some attention. It is calculated as follows,

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} = -\frac{1}{2}\left(2\partial^\mu A^\nu - 2\partial^\nu A^\mu\right) = F^{\nu\mu}, \quad (66)$$

so we altogether have,

$$\partial_\mu F^{\nu\mu} = j^\nu, \quad (67)$$

which is just the contravariant form of (44).

11 The Electromagnetic Stress Energy Tensor

Consider no source charges and the flat spacetime that we have been working with. The electromagnetic stress energy tensor is given as follows,

$$T^{\alpha\beta} = F^{\alpha\mu}F^{\beta}{}_{\mu} - \frac{1}{4}\eta^{\alpha\beta}F^{\mu\nu}F_{\mu\nu}. \quad (68)$$

Let us first comment that it is a symmetric tensor, i.e, $T_{\alpha\beta} = T_{\beta\alpha}$. Also notice that it is not exactly what is obtained if one uses the field theory formalism presented before. (68) can be seen as a part of the divergence-free term derived earlier. Let us see it explicitly,

$$T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\lambda)}\partial^\nu A_\lambda - \eta^{\mu\nu}\mathcal{L} = F^{\lambda\mu}\partial^\nu A_\lambda - \eta^{\mu\nu}\mathcal{L}, \quad (69)$$

$$T^{\mu\nu} = -F^{\mu\lambda}(F^{\nu}{}_{\lambda} + \partial_\lambda A^\nu) + \frac{1}{4}\eta^{\mu\nu}F^{\alpha\beta}F_{\alpha\beta}, \quad (70)$$

$$T^{\mu\nu} = -\widetilde{T}^{\mu\nu} + F^{\lambda\mu}\partial_\lambda A^\nu, \tag{71}$$

where $\widetilde{T}$ is exactly what was in (68). We have to just show that the second extra term has zero divergence.

$$F^{\lambda\mu}\partial_\lambda A^\nu = \partial^\lambda A^\mu \partial_\lambda A^\nu - \partial^\mu A^\lambda \partial_\lambda A^\nu, \tag{72}$$

$$\partial_\mu(F^{\lambda\mu}\partial_\lambda A^\nu) = \partial_\lambda A^\nu \left(-\Box A^\lambda + \partial^\lambda \partial_\mu A^\mu\right) = 0, \tag{73}$$

in the Lorenz gauge. From here on, $\widetilde{T} = T$. Before we look at the components of $T^{\alpha\beta}$, we need the other versions of the Faraday Tensor.

$$F^\beta{}_\mu = \eta^{\beta\alpha}F_{\alpha\mu} = \eta^{\beta\beta}F_{\beta\mu}, \tag{74}$$

where I have abused the diagonality of the Minkowski metric. So the components of this version of the Faraday Tensor are,

$$F^0{}_0 = \eta^{00}F_{00} = 0, \tag{75}$$

$$F^0{}_1 = \eta^{00}F_{01} = (-1)(-E^x) = E^x, \tag{76}$$

$$F^0{}_2 = \eta^{00}F_{02} = (-1)(-E^y) = E^y, \tag{77}$$

$$F^0{}_3 = \eta^{00}F_{03} = (-1)(-E^z) = E^z, \tag{78}$$

$$F^1{}_0 = \eta^{11}F_{10} = (+1)(+E^x) = E^x, \tag{79}$$

$$F^1{}_1 = \eta^{11}F_{11} = 0, \tag{80}$$

$$F^1{}_2 = \eta^{11}F_{12} = (+1)(B^z) = B^z, \tag{81}$$

$$F^1{}_3 = \eta^{11}F_{13} = (+1)(-B^y) = -B^y, \tag{82}$$

$$F^2{}_0 = \eta^{22}F_{20} = (+1)(E^y) = E^y, \tag{83}$$

$$F^2{}_1 = \eta^{22}F_{21} = (+1)(-B^z) = -B^z, \tag{84}$$

$$F^2{}_2 = \eta^{22}F_{22} = 0, \tag{85}$$

$$F^2{}_3 = \eta^{22}F_{23} = (+1)(B^x) = B^x, \tag{86}$$

and the last four values remain the same. So in matrix form we have (where the first index is the row, second is column),

$$F^\beta{}_\mu = \begin{pmatrix} 0 & E^x & E^y & E^z \\ E^x & 0 & B^z & -B^y \\ E^y & -B^z & 0 & B^x \\ E^z & B^y & -B^x & 0 \end{pmatrix}. \tag{87}$$

Ok! Now let's find the complete contravariant form!

$$F^{\alpha\mu} = F^{\alpha}{}_{\nu}\eta^{\nu\mu} = F^{\alpha}{}_{\mu}\eta^{\mu\mu}, \tag{88}$$

$$F^{00} = F^{0}{}_{0}\eta^{00} = 0, \tag{89}$$

$$F^{01} = F^{0}{}_{1}\eta^{11} = E^{x}, \tag{90}$$

$$F^{02} = F^{0}{}_{2}\eta^{22} = E^{y}, \tag{91}$$

$$F^{03} = F^{0}{}_{3}\eta^{33} = E^{z}, \tag{92}$$

$$F^{10} = F^{1}{}_{0}\eta^{00} = -E^{x}, \tag{93}$$

$$F^{11} = F^{1}{}_{1}\eta^{11} = 0, \tag{94}$$

$$F^{12} = F^{1}{}_{2}\eta^{22} = B^{z}, \tag{95}$$

$$F^{13} = F^{1}{}_{3}\eta^{33} = -B^{y}, \tag{96}$$

and hopefully it's clear what pattern is developing. In matrix form,

$$F^{\alpha\mu} = \begin{pmatrix} 0 & E^{x} & E^{y} & E^{z} \\ -E^{x} & 0 & B^{z} & -B^{y} \\ -E^{y} & -B^{z} & 0 & B^{x} \\ -E^{z} & B^{y} & -B^{x} & 0 \end{pmatrix}. \tag{97}$$

Now, we are ready to compute the components of $T^{\mu\nu}$. let us look at its components one by one. First of all we need the quantity,

$$F^{\alpha\beta}F_{\alpha\beta} = -2E^{2} + 2B^{2}, \tag{98}$$

which is readily visible. Here are the components,

$$T^{00} = F^{0\mu}F^{0}{}_{\mu} - \frac{1}{4}(-1)(2)\left(-E^{2} + B^{2}\right), \tag{99}$$

$$T^{00} = E^{2} + \frac{1}{2}\left(-E^{2} + B^{2}\right) = \frac{1}{2}\left(E^{2} + B^{2}\right), \tag{100}$$

$$T^{10} = F^{1\mu}F^{0}{}_{\mu} = B^{z}E^{y} - B^{y}E^{z} = \left(\vec{E} \times \vec{B}\right)^{x}, \tag{101}$$

$$T^{20} = F^{2\mu}F^{0}{}_{\mu} = -B^{z}E^{x} + B^{x}E^{z} = \left(\vec{E} \times \vec{B}\right)^{y}, \tag{102}$$

$$T^{30} = F^{3\mu}F^{0}{}_{\mu} = \left(\vec{E} \times \vec{B}\right)^{z}, \tag{103}$$

$$T^{11} = F^{1\mu}F^{1}{}_{\mu} - \frac{1}{4}(1)(2)(-E^2 + B^2), \tag{104}$$

$$T^{11} = F^{1\mu}F^{1}{}_{\mu} - \frac{1}{2}(-E^2 + B^2) = -(E^x)^2 + (B^z)^2 + (B^y)^2 - \frac{1}{2}(-E^2 + B^2), \tag{105}$$

$$T^{11} = -(E^x)^2 - (B^x)^2 + \frac{1}{2}(E^2 + B^2), \tag{106}$$

and similarly for the other components. In matrix form, we have

$$T^{\alpha\beta} = \begin{pmatrix} U & S^x & S^y & S^z \\ S^x & -\sigma^{xx} & -\sigma^{xy} & -\sigma^{xz} \\ S^y & -\sigma^{yx} & -\sigma^{yy} & -\sigma^{yz} \\ S^z & -\sigma^{zx} & -\sigma^{zy} & -\sigma^{zz} \end{pmatrix}, \tag{107}$$

where the terms appearing are given below,

$$U = \frac{1}{2}(E^2 + B^2), \tag{108}$$

is the energy density in the electromagnetic fields,

$$\vec{S} = \vec{E} \times \vec{B}, \tag{109}$$

is the Poynting vector, and

$$\sigma^{ij} = E^i E^j + B^i B^j - \frac{1}{2}(E^2 + B^2), \tag{110}$$

is the Maxwell stress tensor. Finally, from the divergence freeness of the stress-energy tensor, $\partial_\nu T^{\mu\nu} = 0$, we get the conservation laws,

$$\partial_t U + \nabla \cdot \vec{S} = 0, \tag{111}$$

$$\partial_t \vec{S} - \nabla \cdot \vec{\sigma} = 0. \tag{112}$$

If we consider an infinite volume, the fields vanish at the surface of this infinite volume. Thus $U, \vec{S}$ are conserved; however in a finite volume the conservation laws alone hold.

12 The Lorentz Force on a Charged Particle

We have covered the four Maxwell's equations and how they arise from variational calculus, in addition to the associated stress energy tensor. There is one ingredient missing, and that is the Lorentz force on a charged particle moving in an electric and magnetic field. This is derived from the Lagrangian,

$$\mathcal{L}(x^\mu, u^\mu) = -m\sqrt{u_\gamma u^\gamma} - qA_\gamma u^\gamma, \tag{113}$$

subject to the usual Euler-Lagrange equations,

$$\frac{\partial \mathcal{L}}{\partial x^\mu} = \frac{d}{d\tau}\left(\frac{\partial \mathcal{L}}{\partial u^\mu}\right). \tag{114}$$

Inserting the required derivatives gives us,

$$m\frac{du_\mu}{d\tau} = qu^\gamma F_{u\gamma}, \tag{115}$$

where $u^\gamma = (\gamma, \gamma\vec{v})$. The x spacial component is, for example,

$$m\frac{du_x}{d\tau} = q(\gamma E^x + \gamma v^y B^z - \gamma v^z B^y) = q\gamma(E^x + (\vec{v} \times \vec{B})^x), \tag{116}$$

and taking the γ to the left gives the familiar Lorentz force law,

$$\vec{F} = q(\vec{E} + \vec{v} \times \vec{B}). \tag{117}$$

The time component gives,

$$F_0 = q(\gamma v^x(-E^x) + \gamma v^y(-E^y) + \gamma v^z(-E^z))), \tag{118}$$

$$P_{rad} = m\frac{d\gamma}{dt} = -q\vec{v} \cdot \vec{E} = -\vec{v} \cdot \vec{F}_{electric}, \tag{119}$$

telling that the electric field alone does work on the particle, causing it to radiate its rest mass energy.